# Authoring a Virtual Patient

How an author builds a VP activity in VP Studio

**Literal text**
shown as written

**Generated text**
the LLM writes it

**Interaction**
the agent converses

# From idea to student link

The author's path through VP Studio's simple-mode editor

1

## New case

Start from scratch (Welcome + Final feedback), from a script template, or import a .vp.json file.

2

## Shape the Activity Script

Add, move, rename and remove steps. The editor disables options that would break the script rules.

3

## Fill each phase

Build the phase's pipeline of elements: Literal text, Generated text and Interaction.

4

## Guide the agents

Attach hidden side information and strategies, insert {{fields}}, and tune the agent prompts.

5

## Preview and try

Preview shows what the student sees. Try it plays the draft, with trace notes only the author sees.

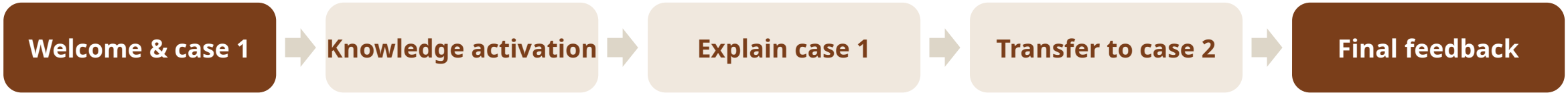
6

## Publish and share

Publish checks readiness and freezes a copy. Students play it at /vp/play/<case id>/.

# The Activity Script

An ordered list of steps; each step after Welcome shows the student a phase




**Fixed ends:** Welcome first, Final feedback last. **Middle:** pick the steps, e.g. the Recall → Explain → Transfer script above.

| Step type | Phase banner | Answer field | Rule |
|---|---|---|---|
| `welcome` | (none) | `{{welcome.answer}}` | Exactly one, first |
| `know` | Knowledge Activation | `{{knowledge.answer}}` | At most once |
| `explain` | Narrative Construction | `{{case1.answer}}` | At most once |
| `assess` | Assessment | `{{assessment.answer}}` | After explain |
| `transfer` | Reasoning Transfer | `{{caseN.answer}}` | After explain; N = 2, 3, … |
| `interview` | Patient Interview | `{{interview.answer}}` | At most once |
| `reflect` | Reflection | `{{reflection.answer}}` | Once |
| `custom` | Its own title | `{{<key>.answer}}` | Unique answer key |
| `final` | (none) | `(none)` | Exactly one, last |

### The case also holds

**`title`**
The activity's name

**`settings`**
Student level (4 / 3 / 2 hints per point), progress bar

**`agentPrompts`**
Tutor persona, patient persona, guardrails

**`strategyPrompts`**
One instruction per strategy used

# Structure of a phase

A step = a phase banner + a pipeline of elements that run in order + the answer it records




## Step keys

**`id · type`** which step

**`title · subtitle`** rename the phase

**`caseN`** transfer: case 2, 3, …

**`answerKey`** custom steps

**`elements`** the pipeline, in order

## At run time

Elements run in order.

A step with no interaction waits for one student reply at its end (except Final).

The answer is that reply, or the student's last message in the interaction.

# Literal text

Shown to the student exactly as the author wrote it

99

## What the author fills in

| | |
|---|---|
| `text` | The exact text the student reads. No {{fields}}: they would show as written. |
| `label` | Short name the author sees in the editor. |
| `card` | Shows the text as a case card with this heading. |
| `ref` | Turns the text into a field, e.g. case1.vignette. |
| `autoPhaseSentence` | Welcome: appends a sentence listing the phases. |
| `reviewMaterials` | Final: appends vignettes, expert explanations, knowledge. |
| `strategies` | Gamification only. Side information may be attached. |

## The student sees, e.g. a case card

**CASE 1**

A 34-year-old woman reports six months of tiredness and shortness of breath when climbing stairs…

## Typical uses

**Welcome message** with the list of phases

**Case vignette** as a card, also a field

**Opening question**: right before an interaction, it lets the student speak first

**Review materials** closing the Final step

# Generated text

The author writes a prompt; the tutor writes the text when the session runs

## What the author fills in

| | |
|---|---|
| `text` | A prompt to the tutor, in the second person, with {{fields}}. |
| `label` | Short name the author sees in the editor. |
| `ref` | Makes the generated text a field for later steps. |
| `isFinalFeedback` | Marks the personal feedback of the Final step. |
| `sideInfo` | Hidden information the prompt can draw on. |
| `strategies` | Final feedback, Gamification. An on-start one (self-rating) asks first. |
| `note` | Author-only reminder, never shown. |

## From prompt to text

**AUTHOR WRITES**

Acknowledge `{{welcome.answer}}` in one sentence without judging it, then introduce a short review of how anemias are classified.

fields filled, tutor persona + guardrails

**TUTOR · WRITTEN WHEN THE SESSION RUNS**

*A text written for this student at this moment, never the same twice.*

## Typical uses

**Phase introductions** that recall earlier answers

**Final feedback** from {{transcript}} and {{evaluator.log}}

# Interaction

A multi-turn conversation between the student and an agent

## What the author fills in

| | |
|---|---|
| `text` | The agent's prompt: what to discuss, what to compare, how to help. |
| `endsWhen` | Optional end condition in plain words, checked by an LLM. |
| `maxTurns` | Required, 1–50: hard limit on student turns. |
| `allowStudentExit` | Shows an I'm done button (on by default). |
| `sideInfo` | Expert explanation (feeds the evaluator), knowledge, hidden findings. |
| `strategies` | Feedback, Patient, Gamification. |
| `label · note` | Editor name and author-only reminder. |

## After each student turn

1. **Student replies**
2. **Hidden evaluator** checks the expert explanation
3. **Turn limit** maxTurns reached?
4. **Ends when…** LLM checks the condition
5. **Agent replies** then back to 1, until it ends

**Who opens:** the student, if an opening question comes just before; else the agent.

**Who the agent is:** the patient in an interview step or with a Patient strategy; otherwise the tutor.

**At end:** at-end strategies get one closing agent turn.

# What every element can carry

Side information, fields and strategies connect the pipeline to the agents

## Side information

Hidden text attached to an element; the student never sees it.

`caseN.expert` expert explanation: the evaluator's checklist

`knowledge` reference text for recall

`case1.findings` revealed by the patient only when asked

Anything else: exam report, rubric, labs

## Fields

{{name}} placeholders filled when the session runs.

`{{case1.vignette}}` authored content, by ref

`{{welcome.answer}}` an earlier step's answer

`{{level.hints}}` system values, also {{transcript}}, {{evaluator.log}}

Never forward: a field must already exist at that step

## Strategies

Reusable instructions, each with a moment: on start, per turn or at end.

**Feedback** hints · direct · exam

**Final feedback** summary · checklist · self-rating

**Patient** forthcoming · worried · reserved

**Gamification** badges